PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Assignment Nine: Diagnose and Prescribe

Assignment

You'll investigate an instance of AI / algorithmic bias, using Narayanan and Kapoor's (2024) frameworks / mechanisms.

1. Begin by finding an instance of bias in AI. You can find one on your own, or you *could* choose one of the following:

   - *Mobley v. Workday* (AI discrimination in hiring)
   - AI Bias in Diagnosing Skin Cancer
   - Gender Bias in AI Decisions Regarding Long-Term Care
   - AI Find Black Women's Hairstyles Less "Professional"
   - Lensa.AI Sexualizes Women, Idealizes Men
   - AI-Driven Age Discrimination in Insurance Coverage
   - Amazon's Recruiting Tool That Disliked Women
   - Racial Bias in Twitter's Image-Cropping Algorithm
   - Bias in AI Tools For COVID-19 Responses
   - Generative AI Bias in Depictions of Aging and Dementia

2. Write a structured analysis of the instance you choose, in two parts:

Part I: Identify the prediction task, the training data, the (potentially, proxy) outcome variable(s), and the feedback mechanism (if any). Then, apply Narayanan and Kapoor's mechanisms framework (discussed in class on Monday, November 9) to explain ***why*** the biases arose in that domain: what was the system trying to predict, what data was it trained on, what proxy problem exists, and if / why simply removing the sensitive attribute wouldn't solve it.

Part II: Proposes a ***specific*** remedy to address the bias in question, and honestly assesses its potential costs and benefits. Be sure to address how your proposed solution addresses the fairness-incompatibility problem from class on Friday.

Details

Your assignment should include a separate title page with your name, the name of the course, the name of the assignment, and the date on which it was completed / submitted. The body of the written assignment should be *double-spaced*, in a conventionally-sized (11 or 12 point) font. Each written part of the assignment should be as short or long as you deem necessary; a rough target

might be a total of 3-4 pages, *not* including references. For this assignment, you should cite any sources you drew upon, consistently using a citation style of your choice.

This assignment must be completed and submitted by Friday, December 4, 2026 at 11:59 p.m. ET. Submit your assignment by *creating / converting your assignment into PDF format*, and emailing it to Dr. Zorn (at `zorn@psu.edu`). Assignments received after the due date will not be accepted. This assignment is worth a possible 100 points.